EAT: A Benchmark for Annotation Transfer with Raw pLM Embeddings

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Most proteins have never been studied in a lab

~150,000,000

protein sequences in UniProt [1]

63,384

proteins with experimentally supported functional annotations [10]

Today: copy the label of the protein with the most similar sequence.
Unreliable once fewer than ~30 % of the amino acids match.

[1] UniProt Consortium. UniProt: the Universal Protein Knowledgebase. Nucleic Acids Res (2025); counts: UniProtKB 2026_03, queried 28.10.2026.
[10] Ribeiro et al. (2020), The FEBS Journal, Table 1.

Approach: Embedding-based annotation transfer (EAT)

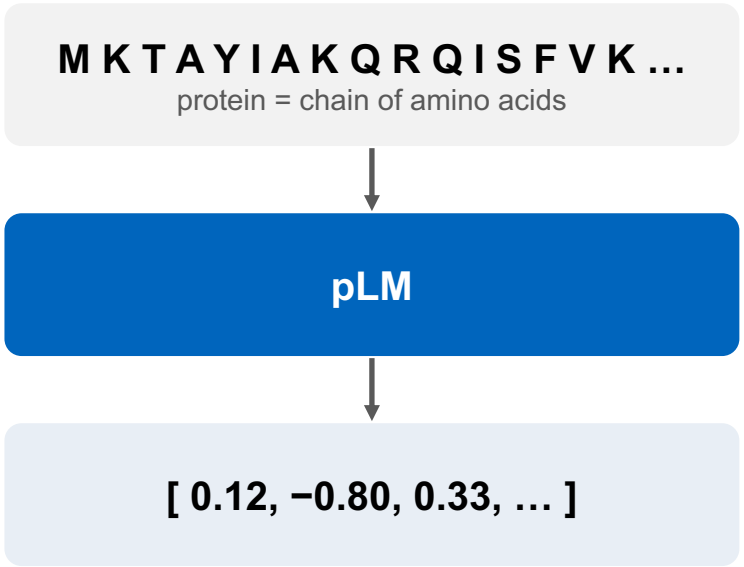


• Without any training, an early version came close to the top 10 of a worldwide prediction contest (CAFA3) [2]

• Embedding-based methods can identify distant structural relationships more accurately than traditional sequence search. [3]

[2] M. Littmann … B. Rost. Embeddings transfer GO annotations beyond homology. Sci Rep (2021).
[3] M. Heinzinger … B. Rost. Contrastive learning on protein embeddings enlightens midnight zone. NAR Genom Bioinform (2022).

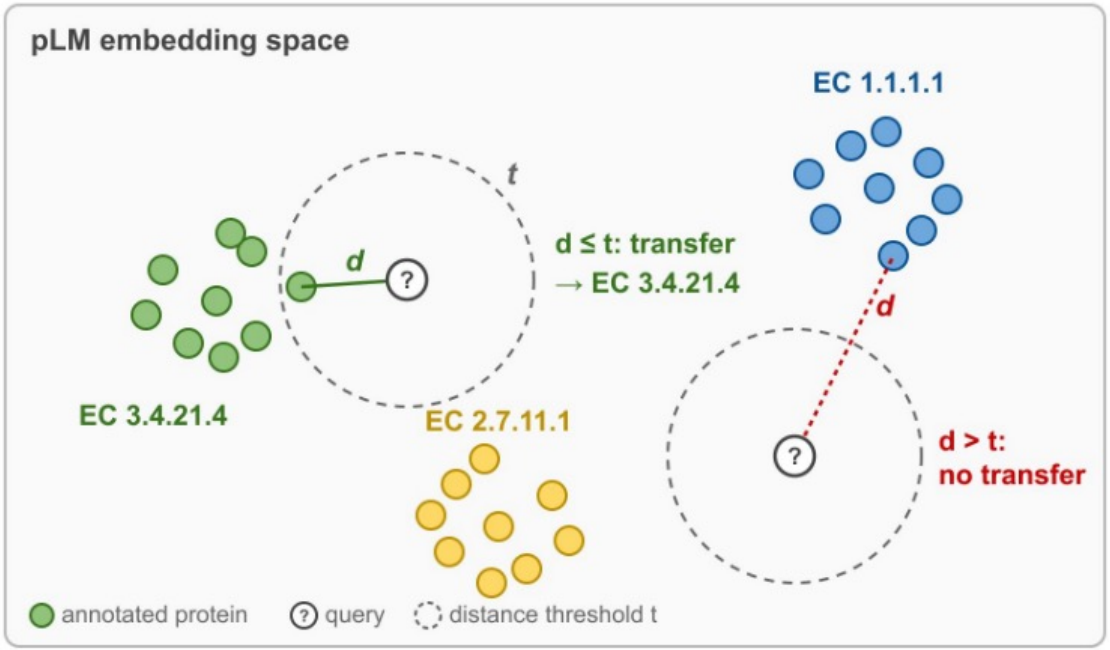
pLMs – Protein Language Models



- pLMs encode proteins into a vector space
- structurally / functionally similar proteins typically end up close together
- different pLMs have different objectives and embedding spaces.

[8] A. Elnaggar … B. Rost. ProtTrans. IEEE TPAMI (2022). [9] Z. Lin … A. Rives. ESM-2. Science (2023). [4] Elnaggar … Rost. Ankh (2023).

Embedding-based Annotation Transfer (EAT)



- **Nearest neighbour:** the closest protein with a known label
- **Distance:** straight line (Euclidean) or angle between vectors (cosine)
- **Threshold:** copy label if below, model-specific

[2] Littmann … Rost (2021).

Labels – CATH and EC

CATH: the 3D shape of a domain

domain = building block of a protein

- **3** Class
- **3.40** Architecture
- **3.40.50** Topology
- **3.40.50.300** Homologous Superfamily

EC: the reaction an enzyme performs

enzyme = protein that speeds up a chemical reaction

- **3** Class
- **3.4** Subclass
- **3.4.21** Sub-subclass
- **3.4.21.4** Enzyme

→ measure accuracy and thresholds per level: up to which level does copying still work?

[6] I. Sillitoe … C. Orengo. CATH: increased structural coverage of functional space. Nucleic Acids Res (2021). [7] IUBMB. Enzyme Nomenclature (ExplorEnz).
Data: [3], [5], UniProt [1].

Research Questions

Q1: Which pLM and which distance measure (e.g., Euclidean or cosine) transfer structure and function annotations best, and where does EAT beat plain sequence similarity?

Q2: Up to which distance is a transfer safe (e.g., ≥95% precision), for structure and for function?
Can distance be turned into a reliability score that means the same for every pLM (e.g., 0.9 = 90% of such transfers are correct)?

Approach

1 Reproduce

Rerun a published result: ProtT5 on CATH and verify

2 Compare

Compare models

Compare EAT vs sequence similarity

3 Homogenize

Both CATH and EC

Distances and thresholds per model

Freeze splits

Lookup set

Tuning set

Test set

[3] Heinzinger … Rost (2022): published ProtT5 result on CATH test219.

References

[1] UniProt Consortium. UniProt: the Universal Protein Knowledgebase. Nucleic Acids Res (2025).

[2] M. Littmann … B. Rost. Embeddings from deep learning transfer GO annotations beyond homology. Sci Rep (2021).

[3] M. Heinzinger … B. Rost. Contrastive learning on protein embeddings enlightens midnight zone. NAR Genom Bioinform (2022).

[4] A. Elnaggar … B. Rost. Ankh: Optimized protein language model unlocks general-purpose modelling. arXiv (2023).

[5] Y. Mahlich … Y. Bromberg. HFSP: High speed homology-driven function annotation of proteins. Bioinformatics (2018).

[6] I. Sillitoe … C. Orengo. CATH: increased structural coverage of functional space. Nucleic Acids Res (2021).

[7] IUBMB. Enzyme Nomenclature. ExplorEnz database, www.enzyme-database.org.

[8] A. Elnaggar … B. Rost. ProtTrans: Toward understanding the language of life. IEEE TPAMI (2022).

[9] Z. Lin … A. Rives. Evolutionary-scale prediction of atomic-level protein structure with a language model. Science (2023).

[10] Ribeiro et al. (2020), The FEBS Journal, Table 1.

Project description: Chair of Bioinformatics (I12), TUM. EAT: A benchmark for annotation transfer with raw pLM embeddings (2026).